MILITARY SPECIFICATION

MANIFOLD, SCUBA, NONMAGNETIC, SINGLE CYLINDER

1. SCOPE

1.1 This specification covers a manifold for use in a single cylinder, U.S. Navy nonmagnetic demand type, self-contained underwater breathing apparatus (SCUBA).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
PPP-B-636 - Box, Fiberboard.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-R-19558 - Regulator, Air, Demand, Diver's
MIL-M-19595 - Magnetic Effect Limits for Non-Magnetic Equipment Used in the Proximity of Magnetic Influence Ordnance.
MIL-R-24169 - Regulator, Air, Demand, Single Hose, Non-Magnetic, Diver's.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810 - Environmental Test Methods.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL BUREAU OF STANDARDS
Handbook H-28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

Compressed Gas Association (CGA)
V-1 - Compressed Gas Cylinder Outlet and Inlet Connections.

(Application for copies should be addressed to the Compressed Gas Association, 11 West 42nd Street, New York, New York 10036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The manifold shall provide a connection to a single aluminum compressed air cylinder and diver's demand air regulator conforming to MIL-R-19558 or MIL-R-24169. It shall consist of a cylinder valve outlet containing a reserve feature and a safety valve. The general arrangement shall be in accordance with figure 1.

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3.1.1 Top and bottom of cylinder. The cylinder described in 3.1 shall be worn on the back of its user with its axis parallel to the axis of the body, the necked end of the cylinder nearest the head. The necked end is the "top" of the cylinder; the other end is the "bottom".

3.1.2 Correct manifold orientation. When the manifold is fitted to the cylinder, a diver's demand air regulator is attached to the manifold, and the whole is worn by a user; the portion of the diver's demand air regulator that is attached directly to the manifold shall lie between the manifold and the user's body.

3.2 Material. All materials used shall be nonmagnetic. The body of the manifold, all metallic parts exposed to unbalanced pressure, and all parts that will likely be immersed in salt water, except the pull rod (see 3.7.1), shall be made of brass.

3.2.1 Magnetic effects. The manifold shall meet the requirements of MIL-M-19595 except that the magnetic effect shall be no greater then 0.05 millioersteds. Upon completion of testing for this requirement (see 4.4.7), each manifold that has successfully met the requirement shall be permanently stamped, etched or engraved with the following symbol:



3.2.2 Safety. All materials used in fabrication or required for operation or maintenance of the manifold shall be safe for use in high partial pressures of oxygen, shall be neither toxic nor noxious, and shall form no toxic or noxious by-products at pressures up to 5000 pounds per square inch (psi) at temperatures up to 250 degrees Farenheit (^{o}F).

3.3 Working pressure. The manifold shall be designed for a working pressure of 3000 pounds per square inch gage (psig).

3.4 Dimensions. Where applicable and except as stated herein, the manifold shall be in accordance with Connection Number 1310 of CGA pamphlet V-1, Valve Outlet Number 1311 and shall include the "O" ring.

3.4.1 Distance. The distance, "T" (see 3.4) shall be not less then 1-1/2 inches nor more than 2-1/2 inches.

3.4.2 Shank. The threaded shank of the manifold shall be in accordance with figure 2.

3.5 Shut-off valve. The manifold shall contain a shut-off valve. With the reserve valve (see 3.7) manually opened and with a supply pressure of 100 psig, the fully opened valve shall pass no less than 6 cubic feet of air per minute to the atmosphere. When fully closed, the valve shall not allow gas to pass with a supply pressure of 3700 psi. The shut-off valve shall turn clockwise to close.

3.6 Safety valve. The manifold shall be fitted with a safety valve that shall actuate when the supply pressure increases to 3700 (± 200) psig. Gas shall escape in the direction away from the user.

3.7 Reserve valve. The manifold shall incorporate a reserve feature of the increased breathing resistance type. When used with the diver's demand air regulator, the reserve feature shall noticeably increase breathing resistance when supply pressure decreases to 300 (+50 - 0) psi. The increased breathing resistance shall be removable by "locking out" the reserve valve by rotating its bonnet approximately 90 degrees in a clockwise direction. When the manifold is oriented in accordance with 3.1.2, the axis of rotation of the reserve valve bonnet shall be perpendicular to the axis of the cylinder, extending to the user's left. Lock-out of the reserve valve shall be accomplished by means of a pull rod, contoured to lie close to the cylinder. Lock-out of the reserve valve shall be accomplished by pulling the bottom end of the rod towards the bottom of the cylinder.

3.7.1 Reserve pull rod. The reserve pull rod shall be a rigid mechanical link made of a suitable metal. Its free end shall terminate in a loop approximately 1 inch in diameter. When pulled, the free end of the rod shall extend to 10 (±2) inches from the bottom of the cylinder measured toward the top of the cylinder. Lock-out of the reserve valve shall require a movement of the pull rod of no less than 1 inch and no more than 2 inches.

3.8 Corrosion resistance. The manifold shall not be subject to malfunction due to corrosion or encrustation of working parts.

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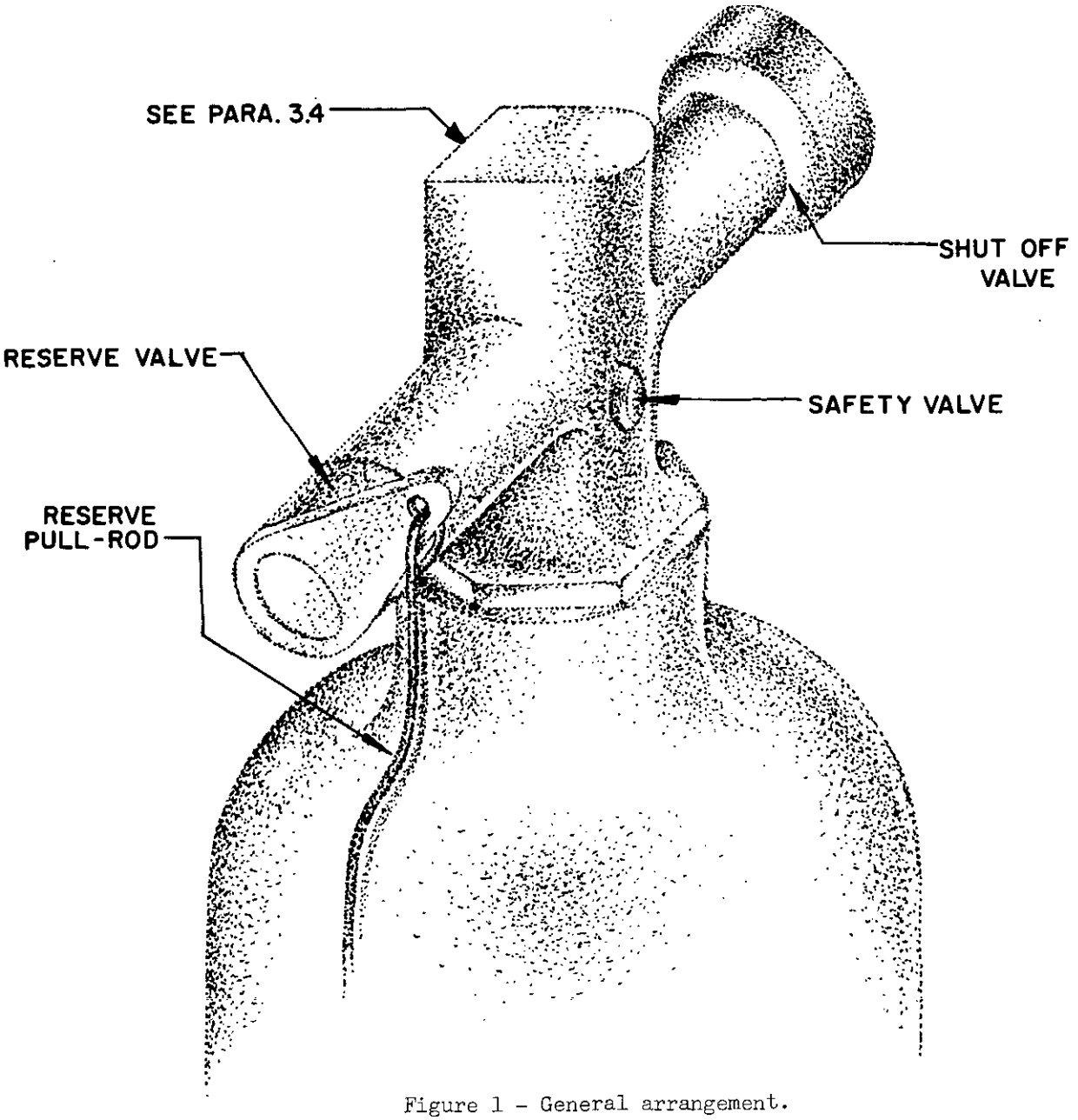
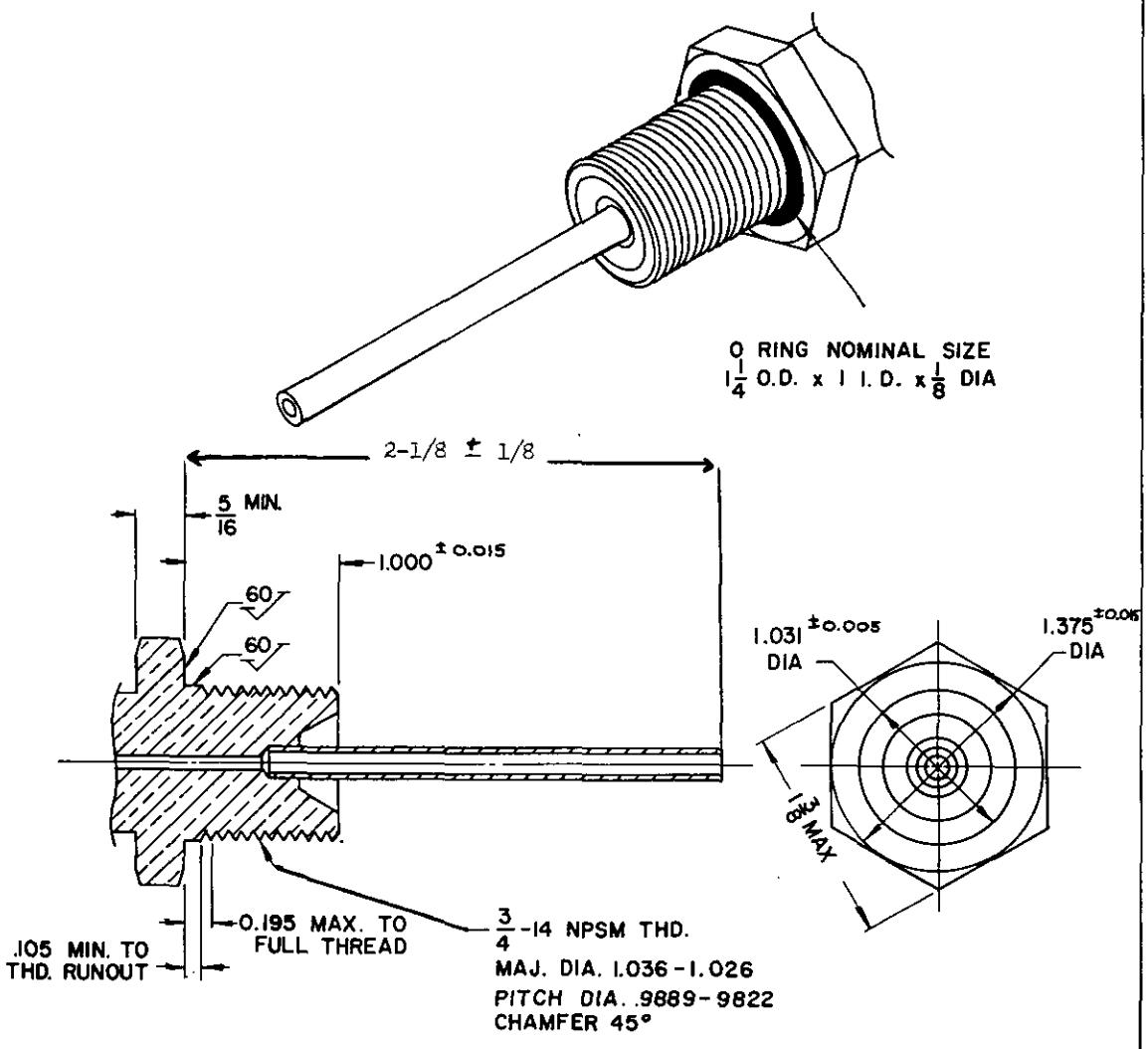


Figure 1 - General arrangement.

SH 9438

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SH 9439

Figure 2 - Threaded shank.

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3.9 Maintainability. All parts of the manifold that are subject to wear through normal use shall be replaceable, using the tools of 3.10.

3.10 Tools. Tools necessary to maintain the manifold shall be furnished preferably in a single, simple, multi-purpose tool which shall meet the magnetic effects requirements set forth in 3.2.1.

3.11 Reliability. The manifold shall have a usage life expectancy based on use cycles of 5000 cycles.

3.12 Finish. All surfaces of the manifold that are visible when the manifold is installed on a cylinder shall be finished with a dull black non-corroding, non-reflective coating that shall neither crumble nor flake.

3.13 Manual. A technical manual in accordance with type II of MIL-M-15071 shall be furnished with each manifold. Three copies of the preliminary manual shall be forwarded for review and approval to the Naval Ship Engineering Center (NAVSEC), Swimming and Diving Branch, Department of the Navy, Washington, D. C. 20360. When ready, three copies of the final manual shall also be forwarded to NAVSEC. Requirements for additional technical manuals shall be as specified (see 6.2).

3.14 Cleaning. The assembled manifold shallbe free from machine chips and foreign particles. All excess lubricant shall be removed and the surfaces shall be free from marks, finger prints and visible accumulations of any sort.

3.15 Workmanship. The finished manifold shall reflect only first class workmanship. There shall be no gross tool markings or surface irregularities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. All manifolds manufactured during the same production run and offered for delivery at the same time shall be considered a lot.

4.2.2 Sampling for examination. From each lot of manifolds, samples shall be selected in accordance with General Inspection Level II of MIL-STD-105 for the examinations of 4.3. The acceptable quality level (AQL) shall be 0.65.

4.2.3 Sampling for quality conformance tests. Except for the tests of 4.4.7 and 4.4.8 (see 4.2.3.1 and 4.2.3.2), samples shall be selected from each lot of manifolds in accordance with Special Inspection Level S-3 of MIL-STD-105 for tests of 4.4.1 through 4.4.6. The AQL shall be 4.0.

4.2.3.1 Magnetic effects sample. Each manifold and each tool in each lot shall be submitted to the magnetic effects test of 4.4.7.

4.2.3.2 Maintainability test sample. A single manifold and sufficient tools shall be selected at random from each lot for the test of 4.4.8.

4.3 Examination.

4.3.1 Visual and dimensional examination. The samples selected as specified in 4.2.2 shall be examined to be in accordance with the manufacturer's drawings and the requirements of this specification. Where conflict between the manufacturer's drawings and the requirements of this specification exists, the requirements of this specification shall govern. Threads shall be checked by means of "GO" and "NO-GO" gauges as specified in H-28.

4.3.2 Cleaning. The samples selected in accordance with 4.2.2 shall be examined for the requirements of 3.14.

4.3.3 Workmanship. The samples selected in accordance with 4.2.2 shall be examined for the requirements of 3.15.

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4.4 Test procedures.

4.4.1 Reserve valve operation. Samples shall be tested to be in accordance with the requirements of 3.7.

4.4.2 Gas flow. Samples shall be tested for the gas flow requirements of 3.5. For this test, the reserve valve shall be installed but "locked out" (see 3.7).

4.4.3 Shut-Off valve. Samples shall be tested to determine that the shut-off valve does not leak gas when subjected to a supply pressure of 3700 psi. The safety valve shall be removed and its receptacle suitably plugged for this test.

4.4.4 Hydrostatic. Samples shall be hydrostatically tested to withstand a supply pressure of 5000 psi. The safety valve shall be removed and its receptacle suitably plugged for this test.

4.4.5 Reliability. Samples shall be tested to deliver 5000 cycles of maintenance-free operation. A cycle of operation shall include the following:

(a) Supply pressure increase to 3000 psi against closed shut-off valve.
(b) Open shut-off valve; supply pressure decreases to point where reserve feature increases breathing resistance.
(c) Reserve feature "locked out" and supply pressure decreases to ambient.
(d) Shut-off valve closed and reserve valve repositioned.

4.4.6 Corrosion resistance. Samples shall be tested for the requirements of 3.8 and 3.12 by exposure to 8 hours of "Salt Fog", Method 509 of MIL-STD-810, followed by 16 hours' exposure to "Sunshine", Method 505 of MIL-STD-810. This combined exposure shall be repeated three times. No rinsing or cleaning shall be accomplished during this test.

4.4.7 Magnetic effects. Each manifold and each tool shall be tested for magnetic effects in accordance with MIL-M-19595 except that the magnetic signature shall not exceed 0.05 millioersteds. Manifolds and tools that do not meet this requirement shall not be offered for delivery.

4.4.8 Maintainability. Utilizing the sample manifold and tool (s) of 4.2.3.2, the manifold shall be dismantled into its component parts and then reassembled. Only the tool(s) supplied in accordance with 3.10 may be utilized. Failure to pass this test shall result in failure of the entire lot.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual orders.)

5.1 Domestic shipment and early equipment use.

5.1.1 Preservation and packaging. Preservation and packaging shall be in accordance with MIL-P-116. No contact preservatives shall be used. The manifold, cleaned in accordance with 3.14, shall be enclosed in a heat sealed plastic bag. A manifold, a set of tools and a manual shall be cushioned and packaged in a snug fitting fiberboard box conforming to PPP-B-636. Box closure and sealing shall be in accordance with the box specification.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal Stock Number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

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5.2.1.1 Preservation and packaging. Preservation and packaging shall be in accordance with 5.1.1.

5.2.1.2 Packing. Packing shall be Level A or B, as specified (see 6.2).

5.2.1.2.1 Level A. Equipment, packaged as specified (see 6.2), shall be packed in fiberboard boxes conforming to PPP-B-636 (class weather-resistant) or PPP-B-640 (class 2) at the option of the contractor. All center and edge seams and the manufacturer's joints shall be sealed and waterproofed with pressure-sensitive tape in accordance with the applicable fiberboard box specification. Box closure and banding shall be in accordance with the applicable box specification. When specified (see 6.2), unit fiberboard containers conforming to class weather-resistant, closed, sealed and banded as specified herein, may be used as the shipping container and need not be overpacked.

5.2.1.2.2 Level B. Units of basic equipment, packaged as specified (see 6.2), shall be packed in fiberboard boxes conforming to PPP-B-636 (class domestic) or PPP-B-640 (class 1) at the option of the contractor. Shipping containers shall be closed and banded in accordance with the applicable container specification, except that fiberboard containers may be sealed with tape conforming to PPP-T-45. When specified (see 6.2), unit containers closed, sealed and banded as specified herein may be used as the shipping container and need not be overpacked.

5.2.1.3 Marking. In addition to any required special markings which shall be specified by the procuring activity (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

6. NOTES

6.1 Intended use. The manifold covered by this specification is intended for use in a demand type diver's self-contained underwater breathing apparatus in a configuration employing a single gas storage cylinder.

6.2 Ordering data. Procurement documents should specify the following:
(a) Title, number, and date of this specification.
(b) Number of extra manuals required (see 3.15).
(c) Packing and marking requirements other than those of 5.1 (see 5.2.1.2).

Preparing activity
Navy-SH
(Project 4220-N125)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 12-66)

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